Excess Deaths in the U.S. During the COVID-19 Pandemic: Differences by Race/Ethnicity, Age, and Sex

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Abstract

This project studies how deaths in the United States changed during the COVID-19 pandemic. Using monthly CDC death counts from January 2019 to June 2021, we compare deaths during the pandemic with the same months of 2019. Our main question is how much deaths went up, and whether the increase was different across race/ethnicity groups. We also compare groups by age and sex. So far, we found that total deaths went up 18.8% in 2020 compared with 2019. The increase was much larger for Hispanic people (44.8%) than for White people (13.9%), and this gap is seen in almost every age group. Deaths from other causes also went up, such as diabetes (16.7%) and Alzheimer's disease (10.5%). Younger adults had a large increase in deaths not from natural causes, such as injuries. Next, we plan to use a linear regression model to test whether the increase during the pandemic was different across race/ethnicity groups.

Introduction

The COVID-19 pandemic changed how many people died in the United States. Many people died from COVID-19 itself, but the pandemic may also have caused more deaths from other causes. For example, some people may have delayed medical care, and others faced more stress and isolation. If we only look at COVID-19 deaths, we could miss these effects.

In this project, we do not focus only on COVID-19 deaths. We compare the number of deaths during the pandemic with the same months of 2019, before the pandemic started. The increase in deaths is called excess deaths. We also look at which causes of death went up, such as Alzheimer's disease and stroke.

The main question that we are focusing on is: compared with the same months of 2019, how much did U.S. deaths go up during the COVID-19 pandemic, and was the increase different across race/ethnicity groups? We also compare groups by age and sex.

Our goal is to test whether the increase in deaths was different across these groups. If some groups had much larger increases than others, this could help public health officials and doctors decide where to focus their efforts in future health emergencies. It can also help researchers understand the effects of the pandemic beyond COVID-19 deaths alone.

Executive Summary

- Total deaths in the U.S. went up 18.8% in 2020 compared with 2019. Every month from March 2020 to June 2021 had more deaths than the same month of 2019.
- The increase was much larger for Hispanic people (44.8%) than for White people (13.9%). The other race/ethnicity groups went up between 26% and 37%.
- This gap is seen in almost every age group, so it is not only explained by differences in age between the groups.

- Deaths from other causes also went up, such as diabetes (16.7%) and Alzheimer's disease (10.5%).
- Younger adults (ages 15 to 44) had an increase of about 23% in deaths not from natural causes, compared with less than 3% for ages 75 and over.

Data

Our data is "AH Monthly Provisional Counts of Deaths for Select Causes of Death by Sex, Age, and Race and Hispanic Origin" from the Centers for Disease Control and Prevention (CDC). We downloaded it from healthdata.gov, and the data is provided by the CDC through data.cdc.gov. It gives the number of deaths in the United States for each month from January 2019 to September 2021. The counts for 2019 are final, and the counts for 2020 and 2021 are provisional, which means they could still change.

The raw data has 3,960 rows and 24 columns. Each row is one month for one group of people, by sex (2 groups), race/ethnicity (6 groups) and age group (10 groups). The columns give the total number of deaths, natural deaths, deaths from 11 groups of causes (such as heart disease, cancer, diabetes, Alzheimer's disease and stroke), and COVID-19 deaths. COVID-19 deaths are given in two columns: deaths where COVID-19 was the underlying cause, and deaths where it was one of multiple causes. We use the underlying cause. The data only gives counts of deaths and does not include population sizes.

We cleaned the data in `11_DataCleaning.Rmd` with these steps:

1. We gave the columns short names.
2. We removed the commas from numbers above 999 and changed the death counts from text to numbers.
3. We changed sex from "M" and "F" in 2019 to "Male" and "Female", so all years match.
4. We made race/ethnicity a factor and age group an ordered factor.
5. We removed four columns we do not need. `analysis_date` and `jurisdiction` have the same value in every row, and `year` and `month` already give the same information as `start_date` and `end_date`.
6. We kept only January 2019 to June 2021. The CDC made this data in October 2021, and deaths with no clear cause went up sharply after June 2021, likely because the cause of many recent deaths was not known yet.
7. We checked for missing values and impossible counts, and found none.

The clean data has 3,600 rows and 20 columns.

Visualization

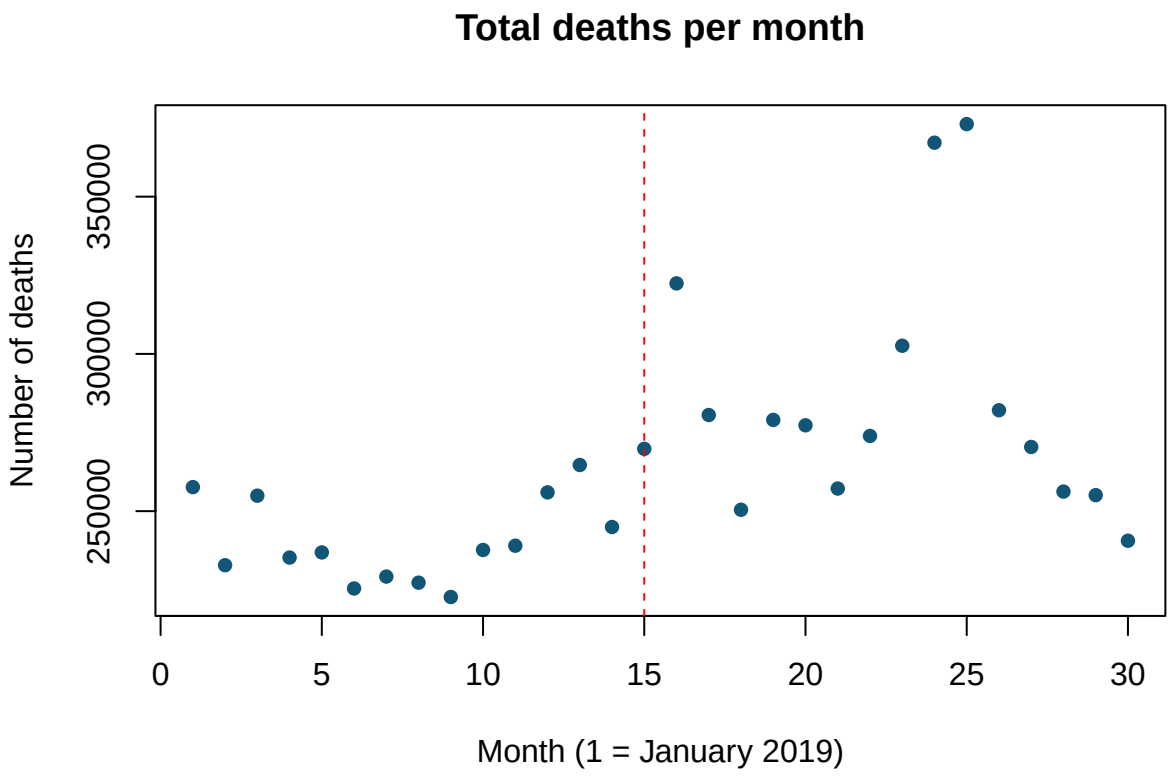
```
library(plot.matrix)
```

```
## Warning: package 'plot.matrix' was built under R version 4.5.3
```

```
source("02_funct_DataProcessing.R")
source("03_funct_Plots.R")
load("22_DataSaving.RData")
```

We first look at the total number of deaths in each month. The dashed red line is at March 2020, when the pandemic started.

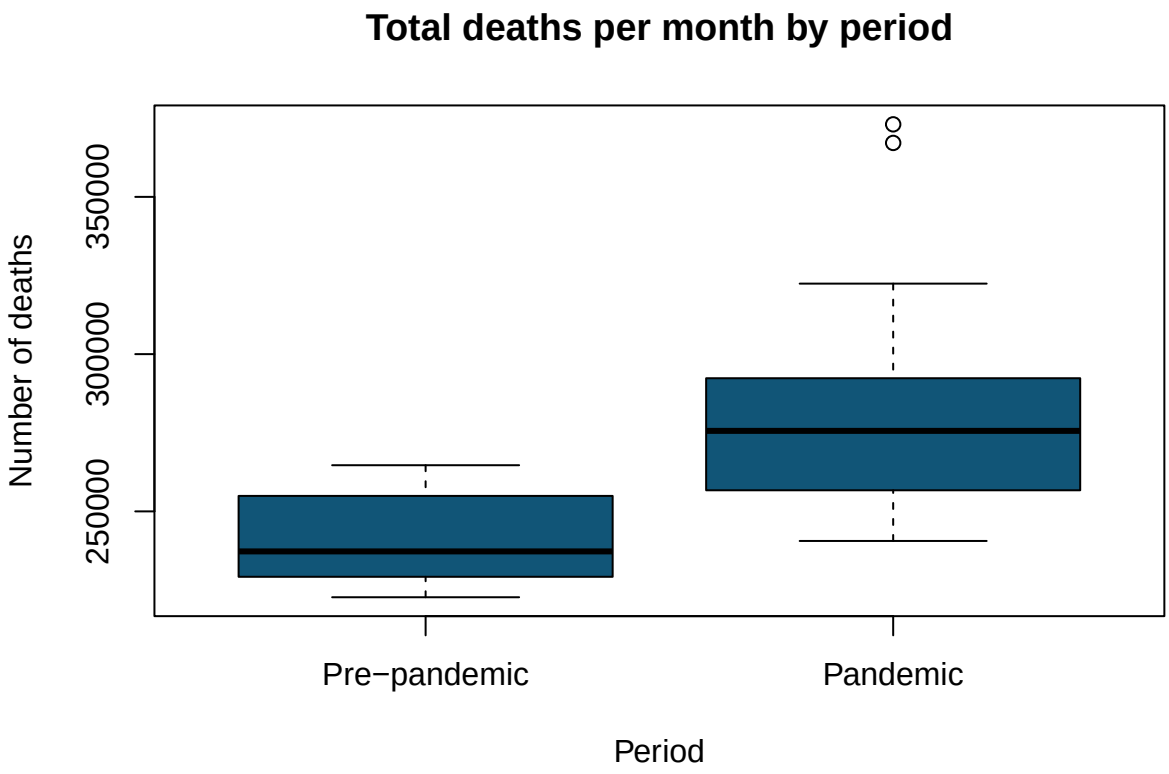
```
plot_monthly_deaths(df_proc, title = "Total deaths per month")
```



Before March 2020, there were between about 223,000 and 265,000 deaths per month. From March 2020 on, every month had more deaths than the same month of 2019, with peaks in April 2020, December 2020 and January 2021. This shows a large increase in deaths during the pandemic.

Next, we compare the number of deaths per month before and during the pandemic.

```
plot_period_boxplot(df_proc, title = "Total deaths per month by period")
```




```
totals <- monthly_totals(df_proc, "all_cause")
summary(totals$deaths[totals$time_index < 15])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  222698  230100  237269  240308  252426  264676
```

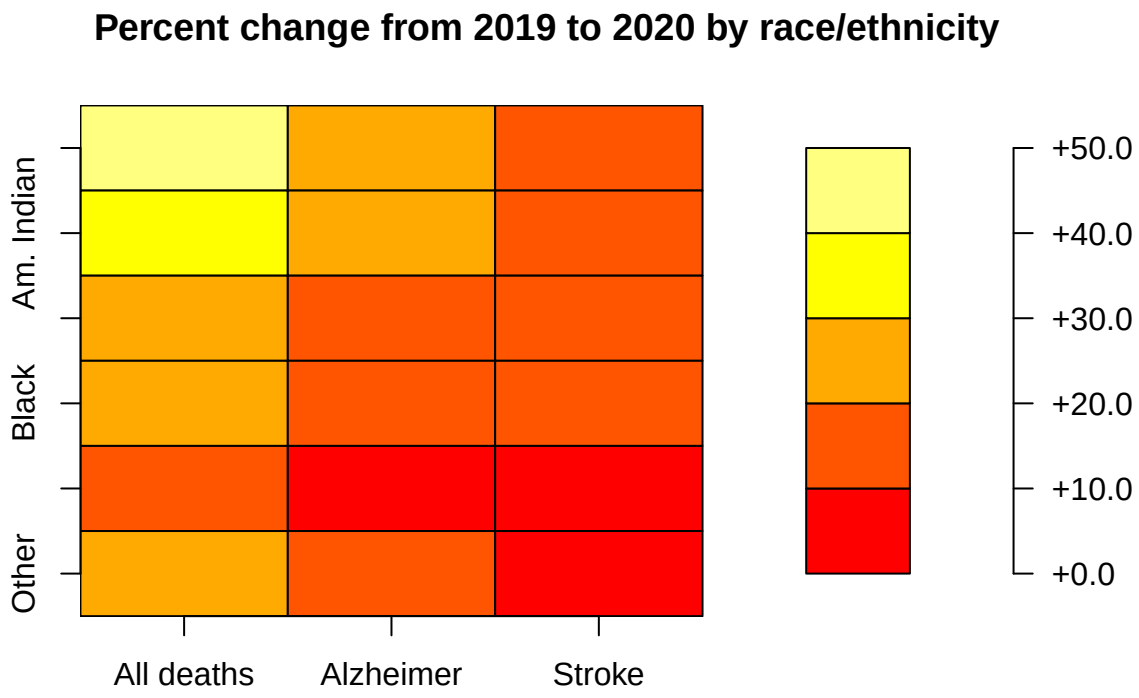
```
summary(totals$deaths[totals$time_index >= 15])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  240591  256949  275599  284863  287213  373060
```

The median number of deaths per month went from about 237,000 before the pandemic to about 276,000 during the pandemic.

To answer our main question, we compare the percent change in deaths from 2019 to 2020 for each race/ethnicity group. Lighter colors mean a larger increase.

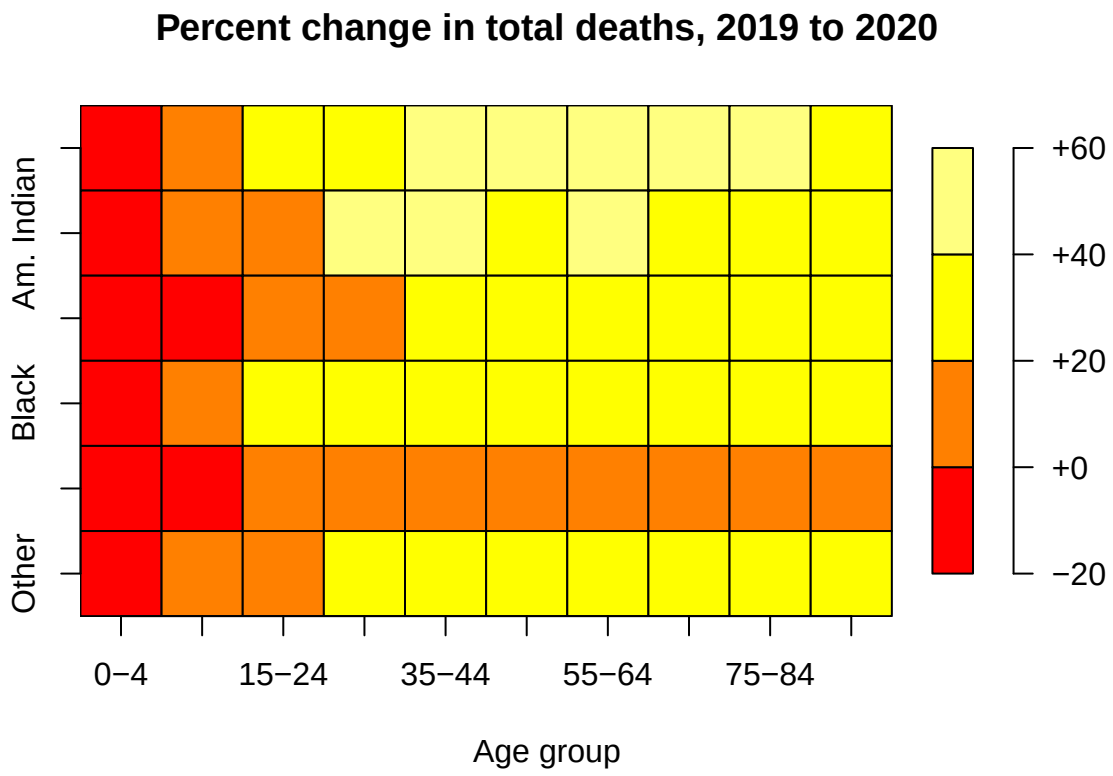
```
plot_change_matrix(df_proc, c("all_cause", "alzheimer", "stroke"),
                   c("All deaths", "Alzheimer", "Stroke"),
                   "race_ethnicity", race_labels,
                   "Percent change from 2019 to 2020 by race/ethnicity")
```



Deaths went up 44.8% for Hispanic people but 13.9% for White people. White people also had the smallest increase in Alzheimer's and stroke deaths. This supports the idea that the increase in deaths was different across race/ethnicity groups.

We also check whether this gap is seen within each age group.

```
plot_race_age_heatmap(df_proc, "all_cause",
                      "Percent change in total deaths, 2019 to 2020")
```



The gap is seen in almost every age group. For example, Hispanic people aged 45 to 84 had increases of about 48% to 57%, while White people of the same ages had increases of about 11% to 16%. So the gap is not only explained by differences in age between the groups.

More plots, including the changes by age group and by sex, are in `31_DataVisualization.Rmd`. For the rest of the project, we plan to add plots of:

- the share of deaths from COVID-19 in each race/ethnicity group
- deaths per month for each race/ethnicity group, to see when the increases happened

Analysis

As of now, we have compared deaths with 2019 using percent changes. Next, we plan to fit a linear regression model for the number of deaths in each month and group. The model will include race/ethnicity, the month of the year (because deaths are higher in winter), and whether the month is during the pandemic. It will also include an interaction between the pandemic and race/ethnicity, which tests whether the increase during the pandemic was different for each group. We also plan to find the correlation between COVID-19 deaths and excess deaths from other causes in each month.

Based on what we see from our results, we expect the increase in deaths to be larger for Hispanic, Black, Asian and American Indian people than for White people. We also expect COVID-19 deaths and excess deaths from other causes to go up at the same times.

Summary of Individual Contribution

Julian Methippara

- Set up the GitHub repository.
- Added the dataset and wrote `00_requirements.R`.
- Wrote the data cleaning functions (`01_funct_DataCleaning.R`) and the data cleaning file (`11_DataCleaning.Rmd`).
- Wrote the report's title, Abstract, Introduction and Data section.
- Knitted the final report to PDF.

Jonathan Methippara

- Wrote the data processing functions (`02_funct_DataProcessing.R`) and the data processing file (`21_DataProcessing.Rmd`).
- Wrote the plotting functions (`03_funct_Plots.R`) and the visualization file (`31_DataVisualization.Rmd`).
- Wrote the report's Visualization and Analysis sections.

Jonathan Waddell

- Added the `README.md` file, with descriptions of each file and steps to reproduce the results.
- Added the report's Executive Summary and this Summary of Individual Contribution.

GitHub Repository

Link: https://github.com/jmethipp2/stat107-pandemic-mortality.git